

THOMAS ZHANG

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TECHNICAL SKILLS

Languages: C/C++, Python, Java, C#, SQL, JavaScript/TypeScript, HTML/CSS

Frameworks: FastAPI, Flask, PyTorch, OpenCV, NumPy, React.js, React Native, Next.js, Node.js

Tools: Git, GitHub Actions, Docker, Docker Compose, AWS (EC2, ALB, EBS), OIDC, Celery, Redis, Apache Kafka, TimescaleDB, PostgreSQL, Supabase, SQLite

EXPERIENCE

Software Engineer Intern

Toronto, ON

Stealth Startup

Jun. 2026 - Current

- Built an autonomous agent harness that ships engineering tickets as tested, mergeable PRs, unattended
- Cut per-issue costs **6x** (\$10+ to <\$3) by A/B testing turn count reduction and per-turn context trimming
- Autonomously resolved **40+ tickets (90%+ success rate)** across two codebases, saving **10+ eng hrs/week**
- Enabled non-engineers to ship real product features by filing tickets the agent resolves directly
- Built a custom sandbox that prevents the autonomous agent from making unsafe or destructive changes
- Gated every change on unit, integration, and e2e tests against a **Docker Compose** replica of production
- Sped up a **Postgres** query **6.5x** (688ms → 106ms) by adding a targeted index, verified on **2M rows**
- Designed a fault-tolerant **SQLite** job queue using lease fencing tokens and a dead letter queue
- Deployed on **AWS EC2** behind an **ALB**, with a snapshotted **EBS** volume and **OIDC**-authenticated deploys
- Added a **GitHub Actions** CI step in the product repo that auto-captures a visual report per PR

Software Developer Intern

Mississauga, ON

Triple J Canada Consulting Inc.

Jun. 2025 - Aug. 2025

- Built a production full-stack tax platform serving **2,000+** clients and **5,000+** returns, replacing paper intake
- Delivered **Flask** backend that reduced staff case prep work by **>30%** by shifting data entry to clients
- Ran live **SQLite** database migrations using Alembic, with **zero loss or downtime**
- Streamlined onboarding with client email verification and admin **RBAC**, eliminating **20+ hrs/week** of work

Software Developer

Waterloo, ON

Watonomous Design Team

Jan. 2026 - Present

- Reduced obstacle costmap input from 307K to ~15K pts/frame via **20×** point cloud downsampling
- Added line-of-sight smoothing to **A*** planner on 60×60 grids, cutting redundant waypoints
- Integrated **YOLOv8** via **ONNX Runtime** on simulated depth camera feeds for obstacle detection

PROJECTS

TradeStream | C++, Python, FastAPI, Next.js, Apache Kafka, Redis, TimescaleDB, Docker

- Architected a platform streaming **1M+ daily ticks** across 100 symbols to a **Next.js** dashboard via WebSockets
- Ran a **C++** microservice computing rolling analytics over a 20-tick sliding window via librdkafka
- Decoupled tick ingestion, analytics, and WebSocket broadcast across three independent **Kafka** consumer groups
- Routed live ticks to **Redis** for recent access and **TimescaleDB** hypertables for historical retention

JobFlow | Next.js, FastAPI, Celery, Redis, Scrappy, Playwright, PostgreSQL, Supabase

- Engineered a job search platform with **FastAPI** serving **150+ concurrent users** via asynchronous processing
- Implemented web-scraping pipeline with **70%+ hit rate** using **Scrappy** with Playwright and rotating proxies
- Designed distributed task queue with **Celery** workers, Redis pub/sub, and WebSocket broadcasting
- Secured API routes with **JWT** authentication and deployed the full stack via **Docker Compose** on Railway

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Software Engineering

Sep. 2024 - Apr. 2028

- Rover Autonomy Developer at Watonomous design team, Percussionist at Orchestra@UWaterloo